

JINNAH INTL, Pakistan

WMO: 417800

Lat: 24.907N

Lon: 67.161E

Elev: 100

StdP: 14.64

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	50.4	53.4	14.4	11.4	72.5	21.0	15.8	72.0	22.7	68.7	19.4	69.9	4.3	30	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.2	102.2	72.8	98.9	73.5	96.9	74.3	82.8	92.1	82.2	91.1	81.6	90.2	9.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.8	160.8	88.0	80.1	156.7	87.6	79.2	151.9	87.1	47.2	92.0	46.4	90.2	45.7	90.7	90.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
21.7	19.1	17.3	DB	45.6	108.2	4.3	2.4	42.5	109.9	40.0	111.3	37.5	112.6	34.4	114.4	
			WB	36.8	86.0	3.3	1.5	34.5	87.0	32.6	87.9	30.7	88.8	28.4	89.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.0	66.9	71.6	79.4	85.3	88.7	89.8	87.5	85.4	85.8	85.0	77.4	69.5
	DBStd	8.26	3.73	4.59	4.26	3.05	2.21	2.35	2.25	2.06	2.53	3.04	3.91	4.01
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	38	24	6	0	0	0	0	0	0	0	0	0	8
	CDD50	11334	524	604	911	1058	1200	1195	1162	1096	1074	1084	821	605
	CDD65	5896	82	190	446	608	735	745	697	631	624	619	371	148
	CDH74	72881	683	1596	4486	7454	10108	10696	9384	7814	7733	7691	3879	1357
	CDH80	36234	75	469	2035	3727	5672	6383	4929	3380	3499	3983	1752	330
Wind	WSAvg	8.2	5.6	6.4	7.0	8.6	11.0	11.7	11.8	10.8	9.4	5.8	4.6	5.1
Precipitation	PrecAvg	6.4	0.3	0.3	0.1	0.0	0.0	0.6	2.1	1.8	0.8	0.1	0.0	0.3
	PrecMax	18.6	3.4	1.3	1.3	1.2	0.2	4.3	10.7	9.5	8.3	1.5	0.3	2.4
	PrecMin	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.1	0.6	0.4	0.3	0.2	0.0	1.1	3.0	2.4	1.8	0.3	0.1	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.5	91.4	100.5	104.4	105.9	105.1	97.6	96.7	102.2	103.6	95.8	88.9
		MCWB	61.5	64.9	68.3	67.4	69.9	78.6	81.1	80.1	76.2	70.4	66.2	63.8
	2%	DB	82.2	87.9	96.2	100.1	100.5	99.4	95.0	92.8	96.7	100.0	93.4	86.1
		MCWB	60.7	64.5	66.7	69.1	74.0	79.8	80.9	79.6	76.0	69.9	66.3	63.4
	5%	DB	80.4	85.7	92.9	96.7	97.0	96.6	93.2	91.0	93.2	96.9	91.4	83.9
		MCWB	60.8	63.4	66.7	70.8	77.4	80.7	80.2	78.7	77.9	69.9	66.1	62.5
	10%	DB	78.4	82.7	89.8	93.6	94.8	94.8	91.5	89.4	91.1	94.7	89.3	81.0
		MCWB	59.6	62.7	67.2	72.5	78.6	80.5	79.8	78.1	77.4	70.8	66.1	61.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.8	72.7	76.1	80.6	83.5	84.5	83.6	82.6	82.5	80.4	76.9	72.5
		MCDB	77.8	82.6	87.5	91.9	94.2	95.5	91.6	90.4	91.1	89.9	85.3	81.4
	2%	WB	67.5	71.0	74.5	79.0	82.1	83.1	82.5	81.1	80.9	79.2	75.0	70.1
		MCDB	75.3	79.5	85.0	89.9	91.8	93.0	90.6	88.1	88.8	88.7	84.0	78.9
	5%	WB	65.6	69.6	73.3	77.9	81.1	82.4	81.8	80.5	79.8	78.3	73.3	68.0
		MCDB	73.9	77.7	83.1	88.0	90.5	91.7	89.6	87.1	87.9	87.6	82.6	77.0
	10%	WB	63.5	68.1	72.0	77.0	80.5	81.7	81.0	79.8	79.0	77.4	71.6	65.7
		MCDB	73.3	76.3	82.0	86.4	89.7	90.8	88.6	86.3	87.3	86.1	81.0	75.6
Mean Daily Temperature Range	5% DB	MDBR	23.0	22.6	21.4	18.6	14.1	11.2	8.7	8.6	11.9	20.7	24.4	24.2
		MCDBR	24.6	25.3	25.5	23.6	18.7	14.4	10.8	11.2	16.1	25.2	25.9	26.4
		MCWBR	12.2	10.5	9.6	8.9	7.7	5.9	4.2	4.3	5.7	9.4	10.7	12.9
	5% WB	MCDBR	21.3	19.2	19.2	16.6	13.3	11.9	9.7	9.5	12.8	16.9	21.9	23.4
		MCWBR	12.8	9.0	8.3	6.8	5.7	5.3	4.5	4.5	5.3	6.9	10.1	13.3
Clear-Sky Solar Irradiance	taub		0.446	0.458	0.513	0.566	0.635	0.733	0.793	0.727	0.621	0.539	0.515	0.486
	taud		2.040	2.041	1.902	1.817	1.715	1.565	1.506	1.605	1.769	1.868	1.881	1.933
	Ebn at Noon		251	259	252	241	224	202	190	202	220	232	226	231
	Edh at Noon		48	51	62	69	76	88	93	84	70	60	56	52
All-Sky Solar Radiation	RadAvg		1393	1710	2045	2276	2351	2160	1752	1712	1873	1794	1477	1321
	RadStd		57	56	56	57	52	102	129	92	101	63	55	51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	-0.86	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page